

Zachary Walters

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SUMMARY

Data Engineer with 10+ years building scalable backend systems and data pipelines across EdTech and FinTech. Currently leading the rebuild of ClassLink's analytics pipeline, an AWS architecture that processes roughly 2 billion events per month into Snowflake. Strong background in AWS infrastructure-as-code, distributed data processing, and AI-assisted engineering workflows that ship production code quickly without sacrificing quality.

TECHNICAL SKILLS

- **Languages:** Python, Go, Java, SQL, JavaScript
- **AWS:** Lambda, Kinesis (Data Streams / Firehose), Step Functions, CDK, S3, DynamoDB, ElastiCache (Valkey), Glue, Redshift, ECS, SQS
- **Data & Storage:** Snowflake, PostgreSQL, SingleStore, Redshift, Apache Spark, MySQL / MariaDB, Oracle RDBMS
- **Practices & Tools:** Infrastructure as Code, CI/CD (GitHub Actions, Jenkins), Docker, Git, LocalStack, AI-assisted development (Claude Code), data quality & observability (CloudWatch)

PROFESSIONAL EXPERIENCE

ClassLink | Remote | December 2025 - Present

Data Engineer

- Re-architected ClassLink's analytics events pipeline (~2 billion events/month), moving from a Firehose-only delivery model to Kinesis Data Streams, Lambda, and S3 with delivery into Snowflake.
- Own all AWS infrastructure as code (CDK): VPC, least-privilege IAM, reactive Kinesis auto-scaling, and multi-stage (dev/staging/prod) deployments.
- Built a Valkey-backed enrichment layer that augments events with roster and demographic data, with fallback to a shared roster data source.
- Designed and built a OneRoster-to-Ed-Fi roster pipeline end to end on AWS Step Functions, with delta syncing and data-quality validation tested against a live Ed-Fi ODS.
- Built a custom AI-assisted engineering workflow on Claude Code that drives a feature from spec to reviewed pull request with automated code review and static-analysis/security gates on every commit; established the team's CI/CD tooling (GitHub Actions, LocalStack) and authored the standards that keep AI-generated code consistent.

Pearson | Iowa City, IA (Remote) | October 2021 - December 2025

Software Engineer

- Engineered concurrent services and APIs in Go for a large-scale assessment platform used across multiple state and international departments of education.
- Optimized ETL pipelines on Redshift and AWS Glue across 50TB+ of assessment data, cutting processing time and cloud cost.
- Designed columnar storage on SingleStore to power fast, live analytics for reporting systems, and extended reporting with new client-requested metrics.
- Built rostering services and integrations that streamlined data flow between systems, and maintained Jenkins CI/CD pipelines.

New Constructs, LLC | Nashville, TN | July 2017 - October 2021

Software Engineer

- Co-designed and implemented complex financial models in Java that powered analysis of public companies for retail and institutional investors.
- Built data aggregation and analytics software, managed large financial datasets, and maintained the flagship monolithic web application alongside CFA-credentialed analysts.

Crystal Clear Fiber, LLC | Nashville, TN | February 2015 - June 2017

Junior Software Engineer

- Built full-stack web applications (PHP, JavaScript), administered AWS VPC infrastructure, and designed and maintained MySQL databases.

EDUCATION

Master of Science (M.S.), Computer Science (Machine Learning & AI focus)

Western Governors University | 2026

Bachelor of Science (B.S.), Computer Science, Minor in Mathematics

Middle Tennessee State University | 2014